Theorem 15: hardness of the optimal-transport edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem16_EditHardness` formalises the reduction of Theorem 15 (p. 12, proof in App. G, pp. 50–51) from Three-Partition. It gives Theorem 4 (1). Shared notions come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Hardness.ThreePartition.dOTEdit_le_iff`: for an instance $(A, a_1, \dots, a_{3m})$ of Three-Partition with all $a_i \geq 2$, the union F_1 of paths on a_i vertices and the union F_2 of m paths on A vertices satisfy

$$d_{\mathrm{ed}}^{\mathrm{OT}}(F_1, F_2) \leq 4m \iff \text{the instance is a YES-instance.}$$

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 15, l. 1600–1602	51	$\|B(F_1, F_2, Q)\|_{(1)} = 2\|E(F_2)\| - 2\|E(F_1)\| + 2\,\mathrm{lost}(Q) = 4m + 2\,\mathrm{lost}(Q)$ for $Q \in \mathrm{DS}$, where $\mathrm{lost}(Q) \geq 0$ is the edge mass of F_1 mapped to non-edges of F_2 (the paper's $\|A^{\pi}_{F_1'} - A_{F_2'}\|_{(1)} = 4mk^2 + 4b$ for blowups)	`GsimOT.Hardness.lost` `Theorem16_EditHardness/Theorem16.lean:57` `GsimOT.Hardness.entrywiseNorm_B_eq` `Theorem16_EditHardness/Theorem16.lean:68` `GsimOT.Hardness.ThreePartition.sum_adj_sub_sum_adj` `Theorem16_EditHardness/Theorem16.lean:113`
Proof of Thm. 15, l. 1611–1616	51	If $\mathrm{lost}(Q) = 0$, no edge of F_1 is mapped to a non-edge of F_2, so every path of F_1 goes into one path of F_2 (the paper's $j(\cdot)$ is constant on paths)	`GsimOT.Hardness.mul_eq_zero_of_lost_eq_zero` `Theorem16_EditHardness/Theorem16.lean:84`
Proof of Thm. 15, l. 1588–1597	50–51	YES-instances: the layout bijection maps edges to edges, so lost $= 0$ and the cost is $4m$	`GsimOT.Hardness.lost_permMat` `Theorem16_EditHardness/Theorem16.lean:93`
Theorem 15, Theorem 4 (1)	4, 12, 50–51	$d_{\mathrm{ed}}^{\mathrm{OT}}(F_1, F_2) \leq 4m$ if and only if the Three-Partition instance is a YES-instance	`GsimOT.Hardness.ThreePartition.dOTEdit_le_iff` `Theorem16_EditHardness/Theorem16.lean:128`

Paper	Page	Statement	Lean declaration(s) and location
Theorem 15 (arbitrary instances), l. 1586–1587	50	The same after adding 3 to each a_i and 9 to A	GsimOT.Hardness.ThreePartition.dOTEdit_shift_le_iff Theorem16_EditHardness/Theorem16.lean:170

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, Sec. 2	2, 5	$C_{\mathrm{ed}}(Q) = \sum_{v,w,v',w'} \lvert D(v,w,v',w')\rvert\, Q(v,w)\, Q(v',w')$ and $C_{\mathrm{ed}}(\pi) = C_{\mathrm{ed}}(P_\pi) = \lVert A_G^\pi - A_H\rVert_{\mathrm{ed}}$	GsimOT.costEdit Common/EditDistance.lean:35 GsimOT.costEditBij Common/EditDistance.lean:39 GsimOT.costEditBij_eq Common/EditDistance.lean:60
Sec. 1, Eq. (1), Sec. 2, App. A	2, 5, 14–15	$d_{\mathrm{ed}} = \min_\pi C_{\mathrm{ed}}(\pi)$ and $d_{\mathrm{ed}}^{\mathrm{OT}} = \min_{Q\in \mathrm{DS}(V,W)} C_{\mathrm{ed}}(Q)$ (infimum, attained); $d_{\mathrm{ed}}^{\mathrm{OT}} \le d_{\mathrm{ed}}$	GsimOT.dEdit Common/EditDistance.lean:42 GsimOT.dOTEdit Common/EditDistance.lean:45 GsimOT.exists_dOTEdit_eq Common/EditDistance.lean:109 GsimOT.dOTEdit_le_dEdit Common/EditDistance.lean:102
Sec. 1, Sec. 2	1–2, 5	$\delta_{\mathrm{ed}} = n^{-2} d_{\mathrm{ed}}$, $\delta_{\mathrm{ed}}^{\mathrm{OT}} = n^{-2} d_{\mathrm{ed}}^{\mathrm{OT}}$ (scaling by $1/n^2$)	GsimOT.deltaEdit Common/EditDistance.lean:48 GsimOT.deltaOTEdit Common/EditDistance.lean:51
Sec. 1, Sec. 2, App. A	2, 5, 14	$C_{\mathrm{ed}}(Q) = \lVert B(G,H,Q)\rVert_{\mathrm{ed}}$ for $Q \ge 0$ (entrywise 1-norm $\lVert\cdot\rVert_{\mathrm{ed}} = \lVert\cdot\rVert_{(1)}$)	GsimOT.entrywiseNorm Common/MatrixNorms.lean:53 GsimOT.costEdit_eq_entrywiseNorm_B Common/MatrixNorms.lean:119

Paper	Page	Statement	Lean declaration(s) and location
Sec. 2, Eq. (3)	5	Birkhoff–von Neumann: $Q = \sum_i \lambda_i P_{\pi_i}$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (also: over all $\sigma \in \mathrm{Perm}(V)$ with $\lambda \geq 0$); conversely every such combination is in $\mathrm{DS}(V, W)$	`GsimOT.exists_birkhoff_decomposition_pos` Common/BirkhoffDecomposition.lean:72 `GsimOT.exists_birkhoff_decomposition` Common/BirkhoffDecomposition.lean:42 `GsimOT.isDS_of_decomposition` Common/BirkhoffDecomposition.lean:23
Proofs of Thms. 15–17	50–57	Disjoint unions of paths (F_1) and of cycles; their degrees and edge counts	`GsimOT.Hardness.pathUnion` Common/PathsAndCycles.lean:56 `GsimOT.Hardness.cycleUnion` Common/PathsAndCycles.lean:226 `GsimOT.Hardness.sum_adj_pathUnion` Common/PathsAndCycles.lean:134 `GsimOT.Hardness.deg_cycleUnion` Common/PathsAndCycles.lean:238
Proof of Thm. 15	50	Three-Partition instances $(A, a_1, \ldots, a_{3m})$; the step of adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition` Common/ThreePartition.lean:26 `GsimOT.Hardness.ThreePartition.IsYes` Common/ThreePartition.lean:39 `GsimOT.Hardness.ThreePartition.shift` Common/ThreePartition.lean:206 `GsimOT.Hardness.ThreePartition.shift_isYes_iff` Common/ThreePartition.lean:223
Proofs of Thms. 15–17	50–56	A solution gives a bijection mapping each path into one component; conversely a $Q \in \mathrm{DS}$ mapping every path into a single component gives a solution (counting mass)	`GsimOT.Hardness.ThreePartition.layoutEquiv` Common/ThreePartition.lean:137 `GsimOT.Hardness.ThreePartition.isYes_of_isDS` Common/ThreePartition.lean:168
Proofs of Thms. 17, 18	54, 57	Attaching $\ell(b)$ leaves to every vertex b of a base graph	`GsimOT.Hardness.addLeaves` Common/Leaves.lean:33

Notes on the formalisation

1. Only the correctness of the reduction is formalised: for the constructed graphs, YES-instances have a small and NO-instances a large distance. The complexity side (polynomial running time, NP-hardness of Three-Partition, Håstad's bound for MaxCut) is not formalised. The results of Section 4 and their shared gadgets are in the namespace `GsimOT.Hardness`.

2. $d_{\mathrm{ed}}^{\mathrm{OT}}$ is `GsimOT.dOTEdit`, the definition Theorem 6 uses. The paper argues with blowups via $d_{\mathrm{ed}}^{\mathrm{OT}} = d_{\mathrm{ed}}^{\bullet}$, cited from [Gro25, Theorem 3.10] (p. 50, l. 1585–1586). The formalisation works directly with doubly stochastic matrices: the same edge count with b/k^2 replaced by $\mathrm{lost}(Q)/2$, where the paper's approximation by $\gamma = 1/(2mA)$ becomes exact because the minimum is attained (`GsimOT.exists_dOTEdit_eq`). So nothing is cited.

3. The formalisation assumes $a_i \geq 2$ (`GsimOT.Hardness.ThreePartition.dOTEdit_le_iff`); the shifted version `GsimOT.Hardness.ThreePartition.dOTEdit_shift_le_iff` holds for every instance.

4. Proposition 57 (p. 57; membership in NP, exact blowup sizes) is not formalised.